CONFIDENT AGENT

Final Technical Report
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Evaluation of a confidence-aware validation pipeline for educational LLM outputs

Primary benchmark: **76 prompts** | Baseline: **llama3.2:3b** | Verifier: **qwen2.5:3b**

Abstract

Large language models (LLMs) can generate fluent but incorrect educational explanations, quiz answers and learning-material recommendations. This project evaluates whether a confidence-aware, agent-based validation pipeline reduces unacceptable hallucinations compared with an unvalidated baseline tutor. We used a fixed benchmark of 76 prompts spanning six categories across COMP200 and COMP301, with obscurity levels from 1 to 3. The baseline generator was llama3.2:3b running locally through Ollama; the Confident Agent reused the baseline response and applied topic-keyed reference notes, a separate qwen2.5:3b verifier, consensus checks, confidence thresholds and external bibliographic/URL checks for materials.

Across all prompts, hallucinations fell from 36/76 (47.4%) to 22/76 (28.9%), an absolute reduction of 18.4 percentage points. The paired exact McNemar test was significant (p=0.0013), with 16 hallucinations prevented and 2 introduced. This result remained significant after excluding five development prompts (32/71 to 19/71; p=0.0023). However, the benefit was concentrated in materials verification: category F fell from 7/15 to 0/15 (p=0.0156). For text-only categories A–E, the rate fell from 29/61 (47.5%) to 22/61 (36.1%), but this did not reach statistical significance (p=0.0654), and was weaker when development prompts were excluded (p=0.1094). The pipeline was approximately 3.8 times slower than the baseline. The findings support a qualified conclusion: external material verification was effective in this benchmark, while the evidence for general text-only hallucination reduction remains inconclusive.

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1. Introduction and research question

LLM-based tutoring systems can explain concepts, generate practice questions and recommend further reading at low marginal effort. Yet fluency is not a reliable signal of correctness. A fabricated API, an incorrect answer key or a nonexistent source can mislead a learner who lacks the subject knowledge to detect the error. In education, the system must therefore do more than produce plausible text: it should assess whether a candidate answer is sufficiently supported before presenting it.

The project investigates a software-engineering intervention rather than attempting to train a new foundation model. The Confident Agent places a validation and decision layer after a baseline generator. It can accept the candidate, regenerate it once with feedback, filter unverified learning materials, or withhold the answer and advise the student to consult course notes or a tutor.

1.1 Research question

Can an agent-based validation architecture detect or prevent unacceptable hallucinations in LLM-generated educational content lesson explanations, quiz answer keys and recommended learning materials compared with a baseline LLM tutor operating without validation?

1.2 Hypothesis and evaluation criteria

The initial hypothesis was that explicit confidence estimation, reference-note support, cross-model checks and external resource verification would reduce hallucinations. Evaluation considers not only the hallucination rate, but also false rejection, correctness, introduced errors, abstention and latency. A system that avoids hallucinations only by refusing most answers would not be a useful tutor.

1.3 Scope

The evaluated domain is undergraduate Computer Science, specifically Java object-oriented programming (COMP200) and software design (COMP301). The benchmark tests answer quality rather than a complete student-facing adaptive-learning interface. The evidence is limited to the particular local model versions, prompts, notes, thresholds and single observed run included in the supplied artefacts.

2. Benchmark design

The final benchmark contains 76 prompts. Each record includes an identifier, category, course, topic, obscurity level, prompt text, ground-truth criteria and scoring guidance. The same prompt set was used for the baseline and the Confident Agent. Ground truth was prepared before the main baseline run, and the benchmark was intended to remain unchanged during comparative evaluation.

Category	n	Purpose
A - Lesson content	18	Conceptual explanations and factual claims
B - Obscure terminology	10	Less common but legitimate concepts
C - Quiz answer key	15	Correctness of answers and reasoning
D - Code reasoning	8	Java behaviour and design reasoning
E - Hallucination traps	10	Prompts involving false or fabricated premises
F - Materials	15	Existence and relevance of recommended resources
Total	76	Six categories; COMP200 and COMP301

2.1 Obscurity and topic coverage

Prompts were assigned obscurity levels 1–3 to examine whether less familiar or more obscure material was associated with higher error rates. The benchmark covers 16 level-1 prompts, 32 level-2 prompts and 28 level-3 prompts. The benchmark also spans 24 COMP200 prompts and 52 COMP301 prompts. This is a curated, course-bounded test set rather than a random sample of all educational queries.

2.2 Ground truth and scoring

Scoring is manual and prompt-specific. The benchmark's ground_truth and scoring fields specify what an answer must establish. Correctness and hallucination are treated as distinct labels, not assumed to be exact complements: an answer may be incomplete or non-hallucinatory without satisfying all correctness criteria. This distinction is important when interpreting the reported counts.

2.3 Materials verification conditions

Category F tests whether suggested books, papers or URLs can be verified. The implemented system uses external checks and a curated resource pool where configured. This category is structurally different from text-only categories because the existence of a resource can often be checked against an external service rather than inferred from generated prose.

3. Baseline evaluation

The baseline was a single-prompt tutor using llama3.2:3b locally through Ollama, with no post-generation validation. The system prompt and generation options were matched to the generator configuration used by the Confident Agent. The baseline used temperature 0, seed 42, CPU inference (num_gpu=0) and a 600-token generation limit, as recorded in the code configuration.

3.1 Baseline result

Measure	Result
Prompts evaluated	76
Hallucinated responses	36/76 (47.4%)
95% Wilson CI for hallucination rate	36.5%–58.4%
Correct responses	37/76 (48.7%)
Hallucinations by category	A 8; B 8; C 2; D 4; E 7; F 7

The baseline hallucination rate was 47.4%, leaving substantial room for improvement. The category counts show that hallucinations were not confined to one prompt type. The highest baseline category rates were in B (8/10, 80.0%) and E (7/10, 70.0%), while quiz answer-key prompts had a lower rate (2/15, 13.3%).

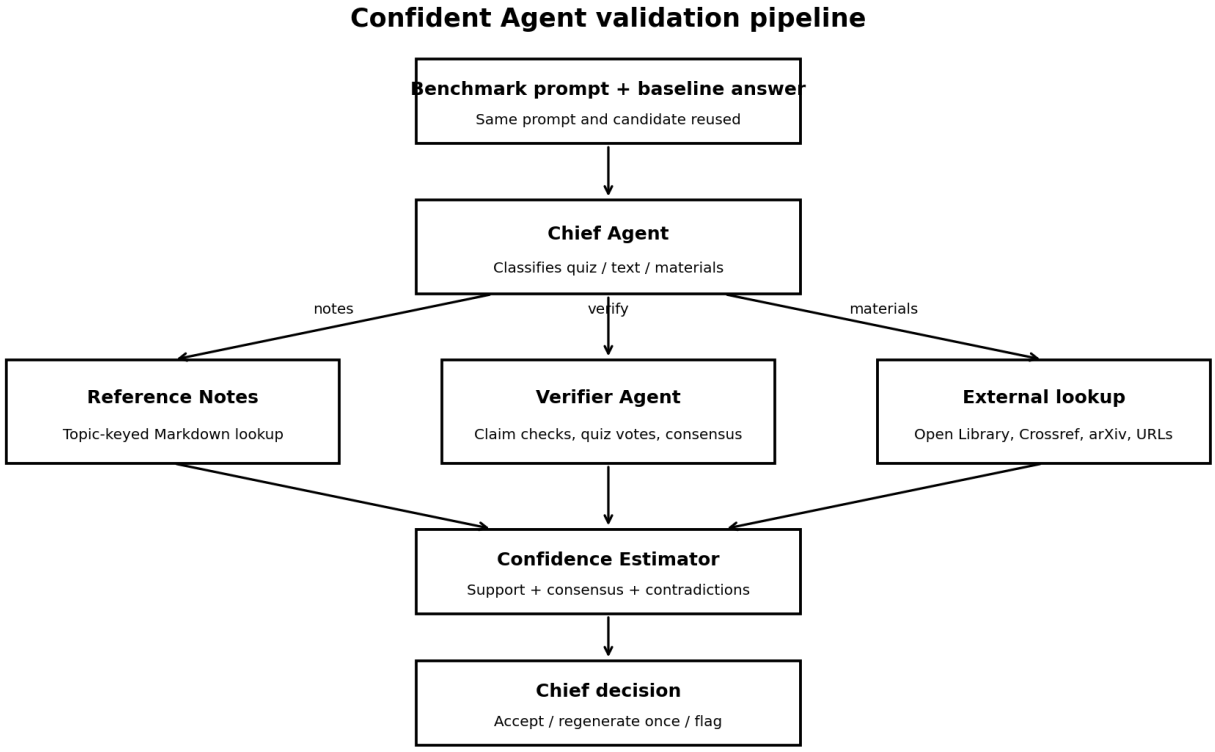
The count of correct answers (37) and the count of hallucinations (36) are separate scoring totals. They do not sum to 76, because the scoring scheme does not require every response to be either fully correct or hallucinated. The corrected Stage 1 figure is 36 hallucinations out of 76, not 37.

3.2 Obscurity pattern

Baseline hallucination rates rose with obscurity: 3/16 (18.8%) at level 1, 15/32 (46.9%) at level 2, and 18/28 (64.3%) at level 3. This descriptive pattern is consistent with obscurity being a useful benchmark dimension, but it should not be interpreted as a causal effect: category composition and prompt wording may also differ across levels.

4. Confident Agent architecture

The system is a Python pipeline coordinated by a Chief Agent. It does not use a general agent framework or vector database. The architecture separates generation from validation and uses a small, bounded set of topic-keyed Markdown reference notes.



4.1 Components

Component	Role in the implemented system
Generator	Produces the original candidate using llama3.2:3b. The baseline candidate is reused rather than silently replacing the baseline run.
Verifier	Uses qwen2.5:3b to assess selected claims, quiz answers and consistency between candidate and independent samples.
Reference notes	Select up to two topic-keyed Markdown notes using keyword/title scoring; notes are capped at 2,800 characters.
External lookup	Checks URLs and bibliographic records through HTTP, Open Library, Crossref and arXiv where relevant.
Confidence estimator	Combines support, consensus and contradiction signals using configured weights and thresholds.
Chief Agent	Classifies the request, applies the decision policy, and returns accepted, regenerated, filtered or flagged content.

4.2 Decision policy and thresholds

For free-text answers, the verifier checks up to five claims. Supported claims contribute 1.0, unsure claims 0.6 and contradicted claims 0.0 to the support score; a contradiction caps confidence at 0.35. Two additional generator samples are used for consensus. The configured initial acceptance threshold is 0.75. If the candidate does not pass, the system may regenerate once using the verifier's concerns and reference notes; the regeneration threshold is 0.60, with no contradictions and at least one checkable claim unless the answer honestly abstains.

Quiz answers use two additional answer votes and a 0.75 agreement threshold. Materials are checked using up to six references, with verified resources filtered into the final response. If the system cannot establish sufficient confidence, it can flag the answer instead of presenting it.

4.3 Design rationale

The architecture uses heterogeneous model families for generator and verifier to reduce dependence on identical generation behaviour, although this does not guarantee independence. Topic-keyed retrieval is simple and auditable for a bounded curriculum, but it can miss semantically relevant notes. External lookup is particularly appropriate for the existence of learning materials, where a URL or bibliographic record can be checked directly.

5. Experimental methodology

5.1 Paired comparison

Each of the 76 benchmark prompts was evaluated under both conditions. The primary outcome was a binary hallucination label for the baseline response and the final surfaced Confident Agent outcome. Because outcomes are paired by prompt, the analysis uses the discordant pairs prompts where the intervention prevented a hallucination versus prompts where it introduced one rather than treating the two sets of responses as independent samples.

5.2 Blinding and scoring

The analysis workflow created shuffled, unlabelled sheets for manual re-scoring. A total of 26 baseline answers were re-scored blind. Agreement was 92.3% for hallucination labels (Cohen's k=0.75) and 88.5% for correctness labels (k=0.60). This supports meaningful, but not perfect, scorer agreement. The group's scoring remains a source of potential bias.

5.3 Development prompts and change control

The development smoke-test prompts were A01, A03, A05, D04 and E02. CHANGES.md records logic and note-selection changes made before the full 76-prompt run, including improved consensus wording, trace logging, a guard against unverified regeneration, and narrower note keywords. The thresholds in config.py were not changed. Results are therefore reported both with and without these five development prompts.

5.4 Reproducibility

The submission bundle includes the benchmark, scored baseline, three blind re-scoring sheets, per-item analysis CSV, full agent results, source code, reference notes, change log, analysis scripts, plots and example traces. The project can be inspected without relying solely on this narrative. Exact output may nevertheless vary if model tags, runtime, external services or source files change.

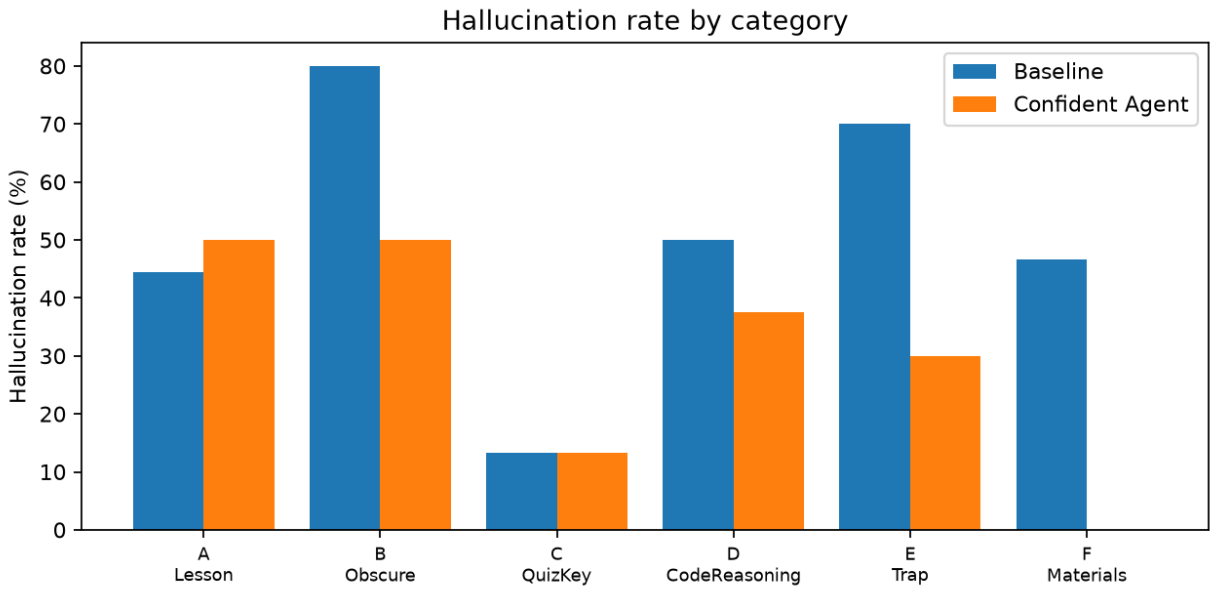
6. Results and comparison

6.1 Primary outcome

Outcome	Baseline	Confident Agent
Hallucinated	36/76 (47.4%)	22/76 (28.9%)
95% Wilson CI	36.5%–58.4%	20.0%–40.0%
Correct	37/76 (48.7%)	41/76 (53.9%)
Flagged/withheld	N/A	9/76 (11.8%)

The absolute hallucination reduction was 18.4 percentage points, corresponding to a relative reduction of 38.9%. The paired bootstrap 95% confidence interval for the absolute reduction was 9.2 to 28.9 percentage points. There were 16 prevented hallucinations and 2 introduced hallucinations.

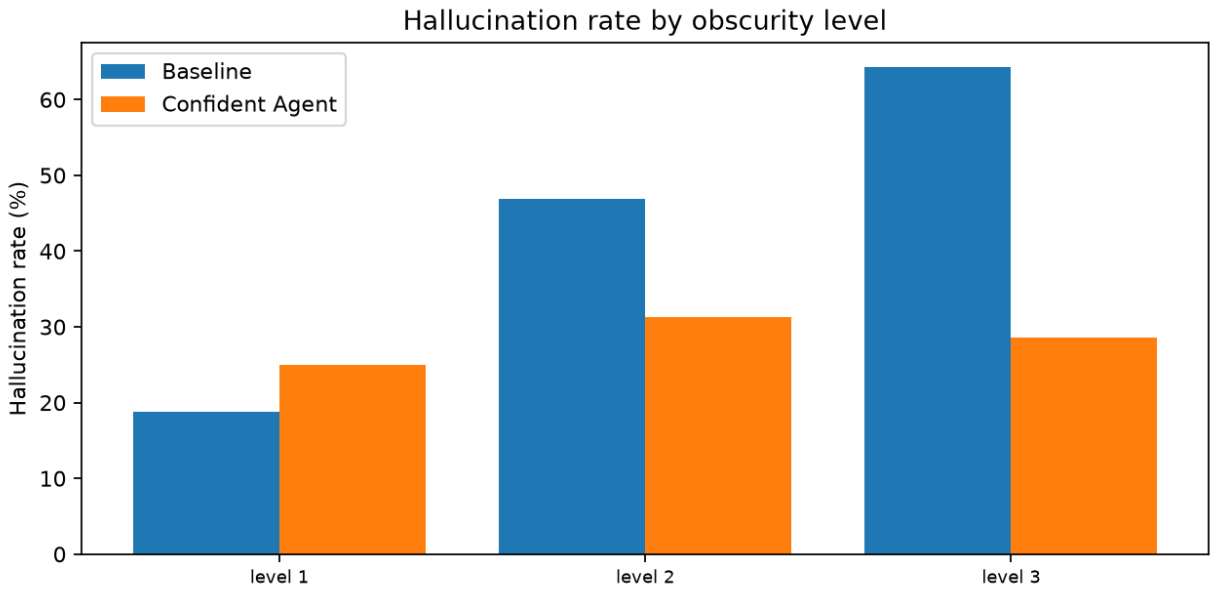
6.2 Category analysis



Category	Baseline HR	Agent HR	Prevented	Introduced
A Lesson	8/18 (44.4%)	9/18 (50.0%)	0	1
B Obscure	8/10 (80.0%)	5/10 (50.0%)	3	0
C Quiz key	2/15 (13.3%)	2/15 (13.3%)	0	0
D Code reasoning	4/8 (50.0%)	3/8 (37.5%)	2	1
E Trap	7/10 (70.0%)	3/10 (30.0%)	4	0
F Materials	7/15 (46.7%)	0/15 (0.0%)	7	0

The largest category-specific improvement occurred in F, where all seven baseline hallucinations were prevented. E also improved descriptively, from 70.0% to 30.0%. Category A increased from eight to nine hallucinations, and D had one introduced hallucination despite two prevented. These mixed category results caution against claiming that the system uniformly improves every kind of educational response.

6.3 Obscurity analysis



For the agent, hallucination rates were 25.0% at level 1 (4/16), 31.2% at level 2 (10/32) and 28.6% at level 3 (8/28). The strong increasing baseline pattern is less apparent after validation. However, the sample is small and category mix differs by obscurity level, so this should be treated as descriptive rather than a confirmed moderation effect.

6.4 Correctness, abstention and error handling

The agent's scored correct-answer count was 41/76 (53.9%). Nine responses were flagged. The reported false rejection rate correct baseline answers withheld was 2/37 (5.4%, 95% CI 1.5%–17.7%). No correct baseline answer was replaced by an incorrect answer according to the reported degradation measure (0/37). Of nine flagged answers, six corresponded to baseline hallucinations, giving a flag precision of 66.7%.

Of the 36 baseline hallucinations, six were withheld, six were repaired into correct responses, four were replaced by non-hallucinatory but not fully correct responses, and 20 remained hallucinated. Thus the pipeline did not eliminate hallucination risk: most baseline hallucinations were still missed.

7. Statistical analysis and robustness

7.1 Paired significance test

The primary paired analysis found 16 discordant pairs in which the baseline hallucinated but the agent did not, and 2 discordant pairs in the reverse direction. The exact McNemar test gave p=0.0013. This is evidence against equal marginal hallucination outcomes for the full benchmark under the paired-test assumptions. It does not establish that the system will perform similarly on other courses or models.

7.2 Confidence intervals and scorer sensitivity

Wilson 95% intervals were 36.5%–58.4% for the baseline and 20.0%–40.0% for the agent. The paired bootstrap interval for the absolute reduction was 9.2–28.9 percentage points. Replacing original baseline labels with the blind re-score labels for the 26 re-scored answers yielded a sensitivity analysis with the same overall point rates and a McNemar p-value of 0.0026.

7.3 Robustness table

Subset	n	Baseline HR	Agent HR	Prevented / introduced	Exact p
All prompts	76	36/76 (47.4%)	22/76 (28.9%)	16 / 2	0.0013
Exclude development prompts	71	32/71 (45.1%)	19/71 (26.8%)	15 / 2	0.0023
A–E, no external lookup	61	29/61 (47.5%)	22/61 (36.1%)	9 / 2	0.0654
A–E, exclude development	56	25/56 (44.6%)	19/56 (33.9%)	8 / 2	0.1094
F materials only	15	7/15 (46.7%)	0/15 (0.0%)	7 / 0	0.0156

The all-prompt effect remains statistically significant when the five development prompts are excluded. In contrast, the A–E text-only effect is directionally favourable but does not meet the 0.05 significance threshold, either with development prompts (p=0.0654) or without them (p=0.1094). The materials-only result is significant in this sample, but it is based on 15 prompts and should not be generalised without replication.

7.4 Latency and computational cost

Measure	Baseline	Confident Agent
Mean seconds per prompt	32.6	124.9
Mean LLM calls per prompt	1	9.3
Relative latency	1.0×	3.8x

The intervention added a mean 92.3 seconds per prompt and increased average LLM calls to 9.3. This is a material usability and compute cost. No monetary cost is reported because the models were run locally and the supplied artefacts do not provide a defensible energy or hardware-cost accounting.

8. Threats to validity

8.1 Construct validity

Hallucination and correctness are manually operationalised using benchmark-specific ground truth. Some responses may contain a mix of correct and incorrect claims, and the binary label compresses severity and scope. The measure does not directly capture pedagogical usefulness, clarity, learning gains or whether a student would be misled.

8.2 Internal validity and development leakage

Five prompts were used in development smoke tests and influenced code or note-selection changes before the full run. This creates a risk of adapting the system to known cases. Reporting results with and without those prompts partially addresses, but does not eliminate, this risk. The benchmark prompts themselves were not changed for the main comparison, and thresholds were not tuned on the 76 prompts.

8.3 Scoring bias and reliability

The group wrote the notes and manually scored outputs, which may favour the system's own assumptions. Blind re-scoring of 26 baseline answers and κ values of 0.75 (hallucination) and 0.60 (correctness) mitigate but do not remove this concern. The sample of re-scored items is limited, and a third independent adjudicator was not used.

8.4 Model and benchmark generalisability

The system uses one 3B generator and a verifier from a different model family. Both can share blind spots, and performance may change with model versions, quantisation, hardware or prompts. The benchmark has 76 prompts concentrated in COMP200/COMP301 and is not representative of all student questions, languages, disciplines or real tutoring sessions.

8.5 Architectural limitations

The verifier cannot reliably judge code semantics; therefore, code-reasoning outputs remain a known weak point. The honest-abstention cues can bypass claim checking, as illustrated by E02, creating a possible route for an answer to receive support without its claims being fully verified. CHANGES.md also notes that UNSURE-only regenerations can be accepted if nothing is contradicted, and three non-material prompts (C09, C13, E08) have no matching note. External services may be unavailable or incomplete, and URL reachability/topic matching does not guarantee source quality.

8.6 Outcome definitions and repeated runs

Withheld answers are counted as non-hallucinated but not correct, which rewards conservative behaviour on the primary hallucination measure. The separate correctness and false-rejection measures are therefore essential. Only one run per prompt was evaluated, so the analysis cannot estimate run-to-run variance. The benchmark size also limits precision in category-specific tests.

8.7 Practical validity

The 3.8x latency penalty may be unacceptable in an interactive tutor without asynchronous processing or a user-visible verification state. Local inference avoids per-token API charges in the measured setup, but it still consumes compute and energy. The current experiment does not quantify these resource costs.

9. Conclusion

On this 76-prompt benchmark, the Confident Agent reduced the observed hallucination rate from 47.4% to 28.9%, with a statistically significant paired result (exact McNemar p=0.0013). The reduction remained significant after excluding the five development prompts. The evidence is not uniform across tasks: materials verification accounted for a substantial share of the gain, while the A–E text-only reduction was not statistically significant. Twenty of 36 baseline hallucinations were missed, two hallucinations were introduced, and the pipeline was 3.8 times slower.

The defensible conclusion is that the architecture is promising for validating the existence of recommended learning materials, but the present experiment does not establish a reliable general-purpose reduction in text-only hallucinations. Future work should add independent scorers, a larger held-out benchmark, repeated runs, code-aware verification, stronger controls around abstention, and explicit latency/energy measurements. A practical deployment should expose uncertainty and verification status rather than presenting the confidence score as a guarantee of correctness.

References

University of KwaZulu-Natal, School of Mathematics, Statistics and Computer Science. (2026). COMP301 Software Design: Confident Agent - Designing for Trust under Uncertainty. Group Project Assignment, Semester Two, Version 1.0.

Dlamini, Y., Ndlovu, S., & Magasela, M. (2026). Confident Agent: An Adaptive Tutoring Agent with Confidence-Aware Content Validation. Project proposal for COMP301 Software Design.

Project artefacts: benchmark.csv; results_baseline_llama3.2_3b_scored.csv; results_agent_full.csv; analysis_results.md; analysis_results_per_item.csv; CHANGES.md; analysis scripts and scoring sheets, included in the accompanying submission bundle.

Appendix guide

The accompanying appendices directory contains the full benchmark, scored baseline, three blind re-scoring sheets, per-item results, full agent outputs and traces, CHANGES.md, robustness output, and the two analysis plots. These are retained as supporting evidence rather than reproducing all 76 long prompts and model responses inside the main report.